

ATM Machine Project Report

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This is a simple command line - Python project showing the working of a simple ATM machine simulation. After card insertion, one can check his/her account balance, deposit and withdraw money in one terminal.

Features

Feature	Description
PIN check	3 attempts allowed.
Checking Balance	Shows the current account balance.
Depositing	Adds money to the account.
Withdrawing	■5000 max per withdrawal.
Daily limit	■25000 max per day, tracked automatically.
Changing PIN	Allows the user to change the PIN.
Exit	Exits the option and "ejects" the card.

Requirements

Python 3.x (no external libraries needed)

Setup & Run

Install Python 3.x. Clone or download this repo. In the project folder, run:

```
python atm.py
```

Usage

Press Enter to insert the card, then enter the PIN (3 tries max). Once logged in, pick an option from the menu and follow the prompts. Choose Exit to quit.

Test Login

Name: Ali Nawaz

Account: XXXX1234

PIN: 1234

Algorithm

1. Start the program.
2. Press Enter to insert the card.
3. Enter the PIN. Allow a maximum of 3 attempts.
4. If the PIN is correct, show the ATM menu.
5. Select balance, deposit, withdraw, change PIN, or exit.
6. For withdrawal, check the withdrawal amount, daily limit, and account balance.
7. Update the balance after a valid deposit or withdrawal.
8. Return to the menu after completing an operation.
9. Choose Exit to eject the card and end the program.

Pseudocode

BEGIN

```
SET account_number = "XXXX1234"  
SET PIN = 1234  
SET attempts = 0  
SET daily_withdrawal = 0
```

PRESS ENTER TO INSERT CARD

```
WHILE attempts < 3  
  INPUT entered_pin
```

```
  IF entered_pin = PIN  
    LOGIN  
    BREAK  
  ELSE  
    attempts = attempts + 1  
    DISPLAY "Invalid PIN"  
  END IF  
END WHILE
```

```
IF login is not successful  
  DISPLAY "Access denied"  
END
```

```
WHILE TRUE  
  DISPLAY MENU  
  INPUT choice
```

```
IF choice = BALANCE  
  DISPLAY balance
```

```
ELSE IF choice = DEPOSIT  
  INPUT amount  
  balance = balance + amount
```

```
ELSE IF choice = WITHDRAW  
  INPUT amount
```

```
  IF amount <= 5000  
    AND daily_withdrawal + amount <= 25000  
    AND amount <= balance
```

```
    balance = balance - amount  
    daily_withdrawal = daily_withdrawal + amount  
    DISPLAY "Withdrawal successful"
```

```
  ELSE  
    DISPLAY "Transaction not allowed"  
  END IF
```

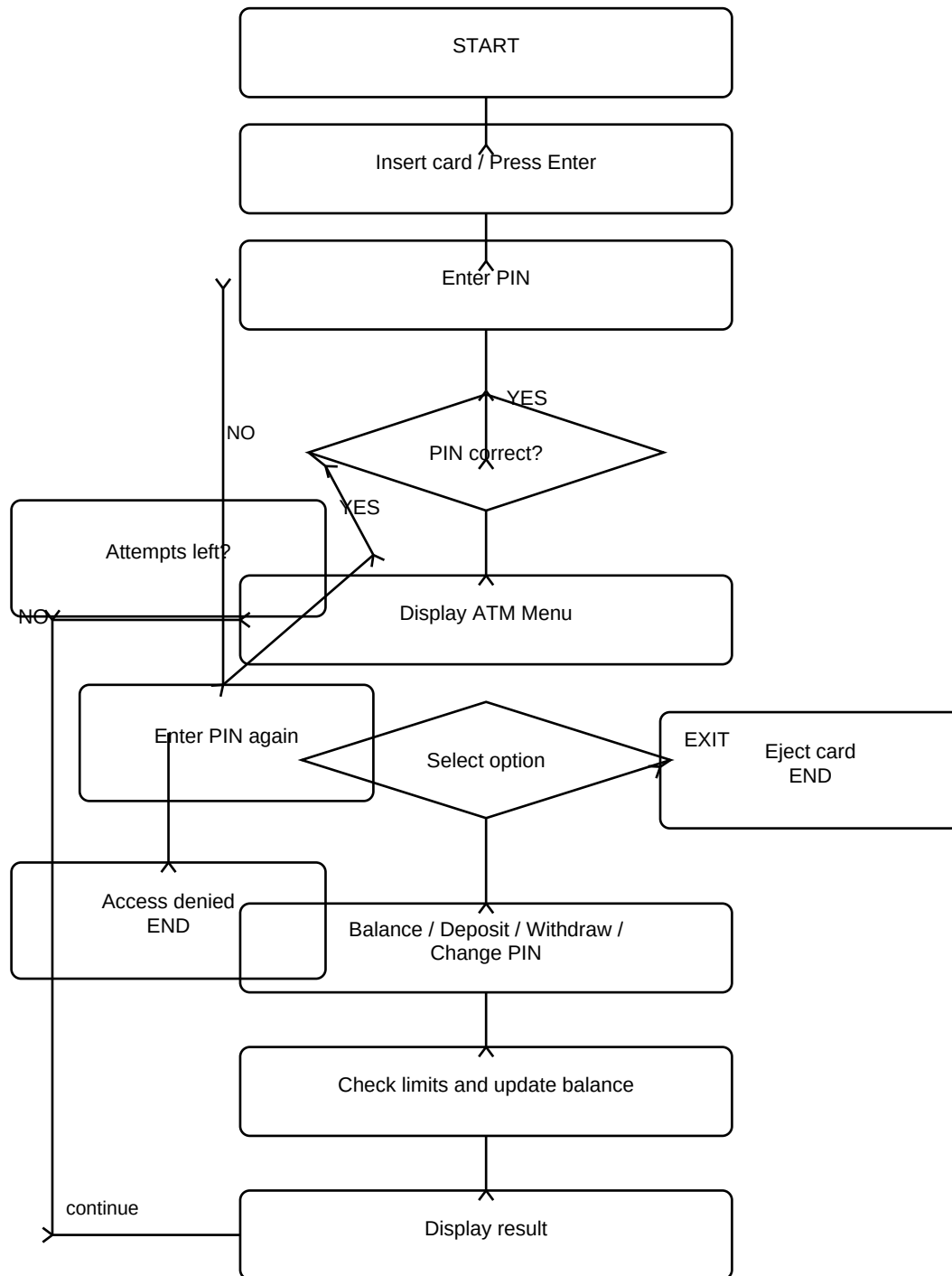
```
ELSE IF choice = CHANGE PIN  
  INPUT old PIN
```

```
  IF old PIN = PIN  
    INPUT new PIN  
    PIN = new PIN  
  ELSE  
    DISPLAY "Invalid PIN"  
  END IF
```

```
ELSE IF choice = EXIT
```

```
DISPLAY "Card ejected"  
BREAK  
END IF  
  
END WHILE  
  
END
```

Flowchart



Project Information

This project is made for a simple ATM machine simulation. It uses preset values for the account number, PIN and account balance. The program runs in one terminal and demonstrates the basic options available after card insertion and PIN verification.